

■ Project Assignment Report: Traffic Signal Synchronization System

Project Title: Traffic Signal Synchronization System Using Chinese Remainder Theorem

Course / Assignment: Number Theory & Discrete Mathematics Application

Team Leadership: Justin Joe Mathew (Team Lead)

■ Does Traffic Signal Synchronization Demo CRT or just LCM/GCD?

Mathematical Distinction:

1. LCM (Least Common Multiple) / GCD:

Used when all signals start at Minute 0 with NO remainder/offset. If Signal 1 (3 min), Signal 2 (5 min), and Signal 3 (7 min) all turn green at Minute 0, the next simultaneous green light is simply $\text{LCM}(3, 5, 7) = 105$ minutes.

2. CRT (Chinese Remainder Theorem):

REQUIRED when signals have DIFFERENT REMAINDERS / OFFSETS.

In our project scenario:

- Signal 1 (3-min cycle) turned green 2 minutes ago $\rightarrow x \equiv 2 \pmod{3}$
- Signal 2 (5-min cycle) turned green 3 minutes ago $\rightarrow x \equiv 3 \pmod{5}$
- Signal 3 (7-min cycle) turned green 2 minutes ago $\rightarrow x \equiv 2 \pmod{7}$

Because the remainders (2, 3, 2) are non-zero and distinct, LCM cannot solve this problem. The system requires the Chinese Remainder Theorem (Extended Euclidean Algorithm & Modular Inverses) to solve for the unique green wave minute ($x = 23$).

■ Conclusion: The project is a genuine demonstration of CRT, specifically showcasing why CRT is needed over simple LCM when real-world cycles start at different offsets.

■ Physical Hardware Prototype & Digital Twin Interface

The project includes both a physical Arduino Nano breadboard circuit and a full digital twin web simulator.

■ Interactive Web App & Project Repository: ``https://github.com/jvstin47/CRT--Chinese-Reminder-Theorem`` (Local Dev Server: ``http://localhost:5173/``)

REAL-TIME SYNCHRONIZATION ENGINE

CRT Traffic Light Lab

Hardware digital twin mirroring an ATmega328P prototype. Demonstrates multi-frequency phase drift and Chinese Remainder Theorem convergence.

Audio Muted

1 Team (10)

Pinout Specs

RUNNING (DRIFTING)

Live Signal Array

ATmega328P View

SIG A3.0s

SIG B5.0s

SIG C7.0s

YELLOW

GREEN

GREEN

Pause

Force Re-Sync

Light Show

Reset

CRT Remainder & Period Combinations:

LCM = 105s

105s

30s

12s

CRT Remainder State Seeker

Solve for timestamp T

A: RED

B: RED

C: RED

Seek to Match

Match occurs at: $t = 8.88s$

CRT Telemetry Engine

105s Recurrence

PERIOD SET

3.0s / 5.0s / 7.0s

HYPERPERIOD LCM

105.00 s

CLOCK PROGRESS

34.55s / 105s

RECURRENCE INDEX

Cycle #3

CRT Modular Recurrence: $t = 0 \pmod{LCM}$

$\Rightarrow LCM(3, 5, 7) = 105$ seconds

$\Rightarrow$ Cycles per LCM: A: 35 | B: 21 | C: 15

$\Rightarrow$ Buzzer chime: Only on synchronization ($t = 0$)

Signal	Remainder ($t \pmod T$)	State
Sig A (3.0s)	1.55 s	YELLOW
Sig B (5.0s)	4.55 s	GREEN
Sig C (7.0s)	6.55 s	GREEN

Multi-Channel Phase Waveform

Window: 0.00s – 105.00s

White cursor sweeps across the 105s hyperperiod. Cyan vertical line marks simultaneous alignment.

SIG A

SIG B

SIG C

Arduino Nano Physical Firmware

Copy Arduino Code

arduino/traffic_light_controller.ino

Signal A: Pin 2 (Red), Pin 3 (Yellow), Pin 4 (Green)

Signal B: Pin 8 (Red), Pin 9 (Yellow), Pin 18 (Green)

Signal C: Pin A5 (Red), Pin A4 (Yellow), Pin A3 (Green)

Buzzer: Pin A1

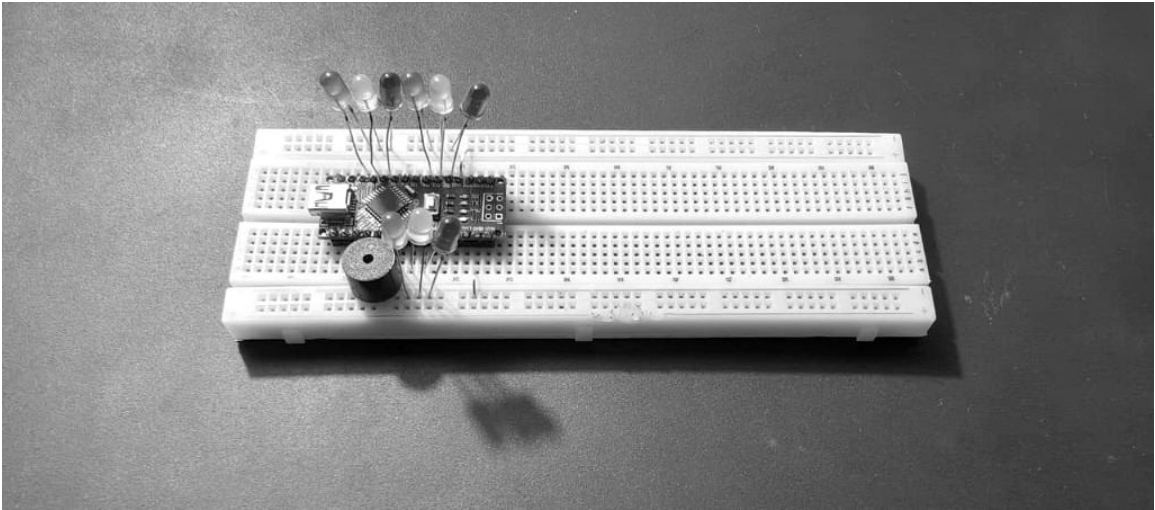
```
const int RED_A = 2;
const int YELLOW_A = 3;
const int GREEN_A = 4;

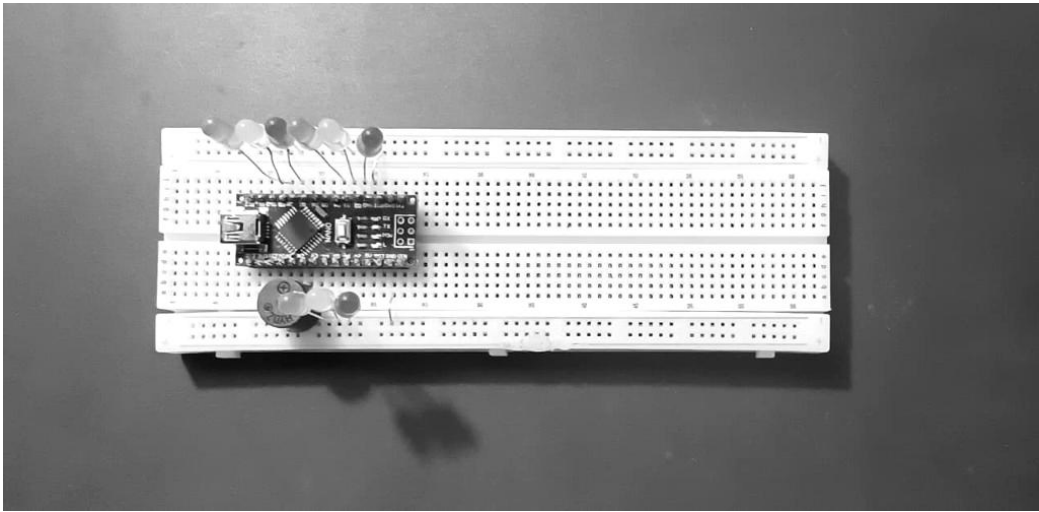
const int RED_B = 8;
const int YELLOW_B = 9;
const int GREEN_B = 18;

const int GREEN_C = A3;
const int YELLOW_C = A4;
const int RED_C = A5;

const int BUZZER = A1;

const unsigned long PERIOD_A = 3000;
const unsigned long PERIOD_B = 5000;
const unsigned long PERIOD_C = 7000;
const unsigned long MASTER_CYCLE = 105000;
```





■ Group Member Roles & Work Division (10 Members)

PROJECT TEAM STRUCTURE		
■ 1. Project Leadership & Model	■ Justin Joe Mathew (Team Lead) & Jyothis Liju	■
■ 2. Hardware Sourcing & Build Testing	■ Joseph Alex & Joseph J	■
■ 3. Web Design & Frontend	■ Johan Geo & Johaan Sam	■
■ 4. Presentation & Showcase	■ Joel Geo Manuel	■
■ 5. Mathematics & Applications	■ Joel Duke, Jyothika Prakash, & Jose Alex	■

■ Detailed Team Member Contribution Breakdown

1. Project Leadership & Working Model Construction

■ Team Members: Justin Joe Mathew (Team Lead) & Jyothis Liju

- Justin Joe Mathew (Team Lead): Guided overall project direction, engineered core system architecture, and led the construction and logic of the working traffic signal model.
- Jyothis Liju: Assisted Justin directly in constructing the working traffic signal model, testing signal synchronization alignment, and verifying physical component interaction.

2. Hardware Sourcing, Build Testing & Application Design

■ Team Members: Joseph Alex & Joseph J

- Joseph Alex: Determined component requirements, purchased hardware parts, executed physical build testing, and contributed to application use-cases.
- Joseph J: Handled hardware part selection, procurement, mechanism assembly testing, and co-developed application integration ideas.

3. Web Design & Frontend Development

■ Team Members: Johan Geo & Johaan Sam

- Johan Geo: Designed the user interface, visual layout, traffic signal dashboard theme, and interactive monitor dials.
- Johaan Sam: Built the frontend components, state integration, responsive layouts, and interactive controls.

4. Presentation & Demo Lead

■ Team Member: Joel Geo Manuel

- Joel Geo Manuel: Handled the presentation structure, demo walkthrough flow, visual slides, and showcase delivery for explaining the project to an audience.

5. Mathematics & CRT Applications Research

■ Team Members: Joel Duke, Jyothika Prakash, & Jose Alex

- Joel Duke: Researched linear congruences and formulated the traffic signal synchronization modulo equations ($3, 5, 7$ minute cycles).

- Jyothika Prakash: Explored real-world CRT applications and authored step-by-step mathematical explanations and hint structures.

- Jose Alex: Investigated Chinese Remainder Theorem proofs, solution bounds ($N = 105$), and modulo remainder properties.